

Project - Deploy and Secure an Azure VM

Objective:

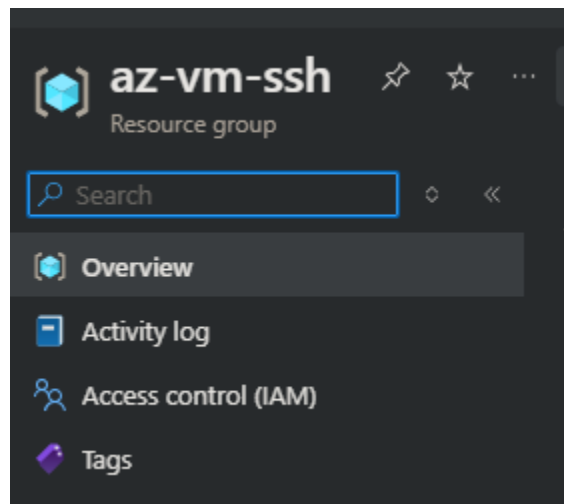
- Create a Basic VM and Secure in Azure

Environment:

- Azure Free Subscription

Recreation steps:

1. Create a resource group in a subscription

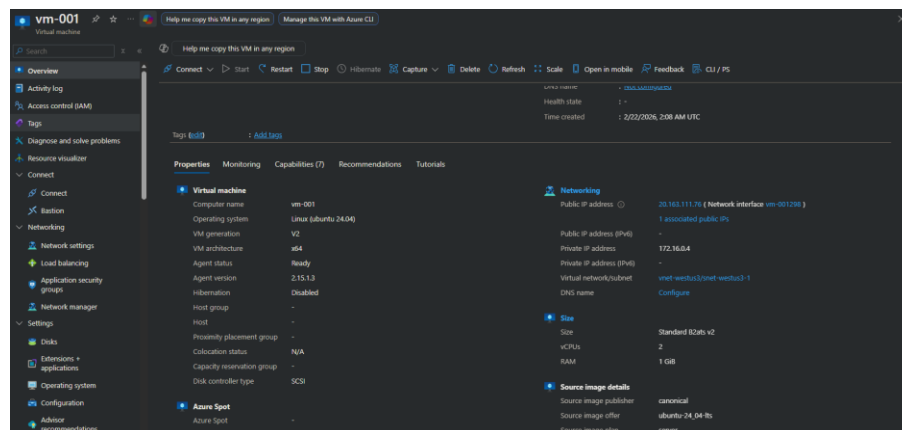


a.

2. Once resource is created -> create a VM

a. This example I used a Linux (ubuntu 24.04)

b. Keeping resource group and VM in same region



c.

3. Next step: Securing NSG to filter traffic to only allow SSH to specific inbound IP (in this case using personal computer to connect only)

SSH
vm-001-nsg

Source ⓘ
IP Addresses

Source IP addresses/CIDR ranges * ⓘ
[Redacted] 137

Source port ranges * ⓘ
*

Destination ⓘ
Any

Service ⓘ
SSH

Destination port ranges ⓘ
22

Protocol
☐ Any
☒ TCP
☐ UDP
☐ ICMPv4
☐ ICMPv6

Action
☒ Allow
☐ Deny

Priority * ⓘ
100 ✓

- a.
- i. Using personal computer IP as source will allow only this ip to connect to VM using ssh according to this rule

4. Connecting to VM via SSH

- a. Using personal computer in CMD
 - i. We can connect to the VM via command
 1. "ssh -i (key) (azure username) @ (VM public IP)"
- b. Once connected we want to secure it

```
Microsoft Windows [Version 10.0.26100.7628]
(c) Microsoft Corporation. All rights reserved.

C:\Users\123817529>ssh -i "C:\Users\123817529\Downloads\vm-001_key.pem" azureuser@20.163.111.76
The authenticity of host '20.163.111.76 (20.163.111.76)' can't be established.
ED25519 key fingerprint is SHA256:DwkLxWnJDwqrb1Axz6In9tjPoyBS+UBr0FKLYLukyos.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.163.111.76' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)
```

i.

5. Running basic updates to Linux Server

- a. `sudo apt update && sudo apt upgrade -y`

```
azureuser@vm-001:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1771 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [328 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
```

i.

6. Configured a basic firewall UFW

- a. `sudo apt install ufw -y`
- b. `sudo ufw allow ssh`
- c. `sudo ufw enable`
- d. `sudo ufw status` (check the status of firewall)

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@vm-001:~$ sudo apt install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
azureuser@vm-001:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
azureuser@vm-001:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y/n)? y
Firewall is active and enabled on system startup
azureuser@vm-001:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
```

i.

7. Disable password authentication

- a. `sudo nano /etc/ssh/sshd_config`

- i. In editor change the below to “no”
 1. PasswordAuthentication no
 2. ChallengeResponseAuthentication no
 3. UsePAM yes
8. Restarting SSH
 - a. `sudo systemctl restart sshd`
9. Security hardening : Install fail2ban
 - a. `sudo apt install fail2ban -y`
 - b. `sudo systemctl enable fail2ban --now`

```
i. azureuser@vm-001:~$ sudo apt install fail2ban -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pyasyncore python3-pyinotify whois
Suggested packages:
  mailx monit sqlite3 python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban python3-pyasyncore python3-pyinotify whois
0 VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@vm-001:~$ sudo systemctl enable fail2ban --now
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
azureuser@vm-001:~$ sudo apt install fail2ban -y
```

10. Enabling Automatic Security Updates
 - a. `sudo apt install unattended-upgrades`
 - b. `sudo dpkg-reconfigure --priority=low unattended-upgrades`

```
i. Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
azureuser@vm-001:~$ sudo apt install unattended-upgrades
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unattended-upgrades is already the newest version (2.9.1+nmu4ubuntu1).
unattended-upgrades set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
azureuser@vm-001:~$ sudo dpkg-reconfigure --priority=low unattended-upgrades
azureuser@vm-001:~$ |
```

Summary:

Created Ubuntu VM using SSH key pair

Applied security controls:

OS updates

UFW firewall enabled

SSH locked to key-auth only

Password login disabled

NSG restricted to my IP

Installed fail2ban

Enabled automatic updates